

AMERICAN LATIN CLASS ATTENDANCE SYSTEM

Requirements

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Requirements

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-01	Public landing page shows academy information, branches, styles, levels, contact/enrollment request.	High	Visitor can open <code>/</code> , read branches/styles/levels and submit an enrollment request.
FR-02	Google OAuth registration/login with internal roles plus controlled manual role login.	High	Backend verifies Google ID token in production, links <code>google_sub</code> , creates app session, supports hashed role-test passwords for manual verification and never trusts Google for app roles.

ID	Requirement	Priority	Acceptance Criteria
FR-03	Secure session management.	High	Session cookie is HttpOnly/SameSite, logout revoke server session, expired sessions return 401, private pages redirect after logout/back button.
FR-04	Role-based and resource-scoped authorization.	High	Student/Teacher/BranchDirector/GeneralDirector/ access is enforced by middleware and resource-scoped policy; BranchDirector access is limited to assigned branches.
FR-05	Branch, student, teacher, class group and class session management.	High	Directors/Admin can create/update core academic authenticated users can list/read.
FR-06	Dance category/style management.	Medium	Urban, Tropical and Ethnic styles are seeded and manageable through API.
FR-07	Student attendance.	High	Teacher/director records one attendance status per student and class session. Duplicates are rejected.
FR-08	Teacher check-in/check-out.	High	Teacher/director opens one active check-in and later closes it for hour calculation.
FR-09	Absence justifications.	Medium	Authorized users submit a justification and director approve/reject it.
FR-10	Scholarship eligibility and evaluation.	High	System marks candidates at $\geq 90\%$ attendance over two months; directors register theory/practice results and approval decision.
FR-11	Level promotion eligibility and evaluation.	High	B1 students can be marked as B2 candidates based on attendance and directors register consistency/theory/practice results.
FR-12	Teacher worked hours and payment report.	Medium	Completed check-in/out records produce hours, hourly rate and amount.
FR-13	Branch and consolidated reports.	Medium	GeneralDirector/Admin can view branch summaries and restricted reports.
FR-14	Audit logs.	Medium	Administrative and attendance actions are recorded and visible to GeneralDirector/Admin.
FR-15	Controlled cache management for safe repeated reads.	Medium	Public, private, no-store and memory-cache policies are explicit; cache hits/misses are visible through headers; state-changing actions invalidate affected cache tags.

Non-Functional Requirements

ID	Requirement	Evidence
NFR-01	REST API versioned with <code>/api/v1</code> .	<code>06Code/src/routes/api.js</code>
NFR-02	Clean Code, MVC, services and repositories.	Controllers delegate to services; persistence is isolated by repository classes.
NFR-03	PostgreSQL normalized schema and seeders.	<code>06Code/migrations/001_initial_schema.sql</code> , <code>06Code/seeders/001_seed.sql</code> , <code>06Code/scripts/seed-role-test-data.js</code>
NFR-04	No production secrets in repository.	<code>.env.example</code> contains placeholders plus clearly temporary local seed defaults that must be rotated/removed before production.
NFR-05	Restricted CORS.	<code>CORS_ORIGINS</code> env configuration.
NFR-06	Auth rate limiting.	<code>express-rate-limit</code> on <code>/auth/google</code> .
NFR-07	Consistent JSON responses.	<code>ApiResponse</code> success/error envelope.
NFR-08	Centralized validation.	Zod validators under <code>src/validators</code> .
NFR-09	Consistent error handling.	<code>AppError</code> and <code>errorHandler</code> middleware.
NFR-10	AWS deployment preparation.	EC2/RDS/ALB guide and PlantUML deployment diagram.
NFR-11	Production-safe configuration.	Production/staging fails fast on default secrets, missing database URL or enabled mock Google tokens.
NFR-12	Controlled HTTP and in-memory cache behavior.	<code>03Documentation/cache-management.md</code> , <code>06Code/src/services/CacheService.js</code> , <code>06Code/src/middleware/cacheControl.js</code> , <code>070ther/nginx-alc-frontend.conf</code>

User Stories

ID	Story
US-01	As a Visitor, I want to request enrollment from the public page so the academy can contact me.
US-02	As a Student, I want secure access to my private academic system session.
US-03	As a Teacher, I want to record student attendance and check in/out for my worked hours.
US-04	As a BranchDirector, I want to manage students, teachers, classes and absence justifications for my branch.
US-05	As a BranchDirector, I want to review scholarship and promotion candidates before registering evaluations.
US-06	As a GeneralDirector, I want consolidated branch reports and audit logs.
US-07	As an Admin, I want to assign internal roles, branch access and role-test users without depending on Google profile data.
US-08	As a school director, I want repeated catalog and report reads to be faster while private academic data remains protected from browser or shared-cache reuse.

Role/Permission Matrix

Feature	Visitor	Student	Teacher	BranchDirector	GeneralDir
View landing page	X	X	X	X	
Submit enrollment request	X				
Login/logout/session		X	X	X	
List own/private data		X	X	X	
Record student attendance			X	X	
Teacher check-in/out			X	X	
Submit absence justification		X	X	X	
Review absence justification				X	

Feature	Visitor	Student	Teacher	BranchDirector	GeneralDir
Manage academic data				X	
Scholarship/promotion evaluation				X	
Branch reports				X	
Consolidated reports					
Audit logs					
Assign roles					
Assign branch access					
Use temporary seeded role login		X	X	X	

Business Rules

- BR-01: Scholarship candidate requires at least 90% attendance in a two-month period.
- BR-02: Scholarship approval is never automatic; directors must register theory and practice evaluation results.
- BR-03: Scholarship percentages allowed are 25%, 50%, 75% and 100%.
- BR-04: Level promotion applies to B1 students moving to B2.
- BR-05: Level promotion approval requires attendance evidence, consistency score, theory score and practice score.
- BR-06: A student can have only one attendance record per class session.
- BR-07: A teacher can have only one open check-in at a time.
- BR-08: Roles are owned by the application and are not inferred from Google claims.
- BR-09: BranchDirector permissions require explicit branch assignment and must not expose other branches.
- BR-10: Student and Teacher users can only access their own academic/profile records and teaching context.
- BR-11: Private pages and APIs must use `Cache-Control: no-store`.
- BR-12: Administrative and academic state-changing actions must be audited.
- BR-13: Manual role-test credentials must be stored as password hashes and remain disabled unless `POSTMAN_LOGIN_ENABLED=true`.
- BR-14: Cacheable non-sensitive responses must declare an explicit TTL, expose cache evidence headers and invalidate affected memory-cache tags after state-changing academic actions.

Prioritized Backlog

1. Auth/session, RBAC, no-store private pages and tests.
2. PostgreSQL schema, seeders and repository abstraction.
3. Branch/student/teacher/style/class management APIs.
4. Student attendance and teacher check-in/check-out APIs.
5. Absence justifications and audit logs.
6. Scholarship and level promotion candidate/evaluation workflows.
7. Reports by branch, teachers and consolidated roles.
8. Landing page, enrollment form and private dashboard shell.
9. AWS EC2/RDS/ALB deployment documentation and diagrams.
10. Controlled cache management with automated evidence.
11. Final test evidence and requirement reevaluation.

Requirement-To-Endpoint Traceability

Requirement	Endpoints
FR-01	GET / , POST /api/v1/enrollment-requests
FR-02	POST /api/v1/auth/google , POST /api/v1/auth/login , GET /api/v1/users , PATCH /api/v1/users/{id}/role
FR-03	GET /api/v1/auth/me , POST /api/v1/auth/logout , /private/dashboard.html
FR-04	All protected endpoints through requireAuth / allowRoles
FR-05	/api/v1/branches , /api/v1/students , /api/v1/teachers , /api/v1/class-groups , /api/v1/class-sessions
FR-06	/api/v1/dance-categories , /api/v1/dance-styles
FR-07	POST /api/v1/student-attendance
FR-08	POST /api/v1/teacher-attendance/check-in , PATCH /api/v1/teacher-attendance/{id}/check-out
FR-09	/api/v1/absence-justifications , /api/v1/absence-justifications/{id}/review
FR-10	GET /api/v1/reports/scholarships/{studentId}/candidate , /api/v1/scholarship-evaluations
FR-11	GET /api/v1/reports/level-promotions/{studentId}/candidate , /api/v1/level-promotion-evaluations
FR-12	GET /api/v1/reports/teachers/{teacherId}/payment

Requirement	Endpoints
FR-13	GET /api/v1/reports/branches/summary
FR-14	GET /api/v1/audit-logs
FR-15	GET /api/v1/auth/config , GET /api/v1/roles , GET /api/v1/branches , GET /api/v1/reports/branches/summary , static /assets/* , protected /api/analytics/v1/*